Longitudinal Data Analysis with R Statistical Software

Day 3: Mixed-effects models

Benjamin French and Bryan Shepherd

Vanderbilt-Nigeria Biostatistics Workshop

April 27 – May 1, 2026

Day 2 review

Mixed-effects models

⋆ Contrast outcomes both within and between individuals

- Assume that each subject has a regression model characterized by subject-specific parameters; a combination of
  - Fixed-effects parameters common to all individuals in the population
  - Random-effects parameters unique to each individual subject
- Although covariates allow for differences across subjects, typically cannot measure all factors that give rise to subject-specific variation
- Subject-specific random effects induce a correlation structure

(Laird and Ware, 1982)

Set-up

For subject i the mixed-effects model is characterized by

$$y_i = \{y_{i1}, y_{i2}, \ldots, y_{im_i}\}^\mathsf{T}$$

$$\begin{aligned}
\beta &= \{\beta_0, \beta_1, \beta_2, \ldots, \beta_p\}^\mathsf{T} && \text{Fixed effects} \\
x_{ij} &= \{1, x_{ij1}, x_{ij2}, \ldots, x_{ijp}\} && \\
X_i &= \{x_{i1}, x_{i2}, \ldots, x_{im_i}\}^\mathsf{T} && \text{Design matrix for fixed effects}
\end{aligned}$$

$$\begin{aligned}
b_i &= \{b_{i0}, b_{i1}, b_{i2}, \ldots, b_{iq}\}^\mathsf{T} && \text{Random effects} \\
z_{ij} &= \{1, z_{ij1}, z_{ij2}, \ldots, z_{ijq}\} && \\
Z_i &= \{z_{i1}, z_{i2}, \ldots, z_{im_i}\}^\mathsf{T} && \text{Design matrix for random effects}
\end{aligned}$$

for $i = 1, \ldots, n$; $j = 1, \ldots, m_i$; and $q \leq p$

Linear mixed-effects model

Consider a linear mixed-effects model for a continuous outcome y_{ij}

1. Model for response given random effects

$$y_{ij} = x_{ij}\beta + z_{ij}b_i + \epsilon_{ij}$$

   with

   - x_{ij}: vector a covariates
   - β: vector of fixed-effects parameters
   - z_{ij}: subset of x_{ij}
   - b_i: vector of random-effects parameters
   - ϵ_{ij}: observation-specific measurement error

2. Model for random effects

$$\begin{aligned} b_i &\sim N(0, D) \\ \epsilon_{ij} &\sim N(0, \sigma^2) \end{aligned}$$

   with b_i and ϵ_{ij} assumed to be independent

Choices for random effects

Consider the linear mixed-effects models that include

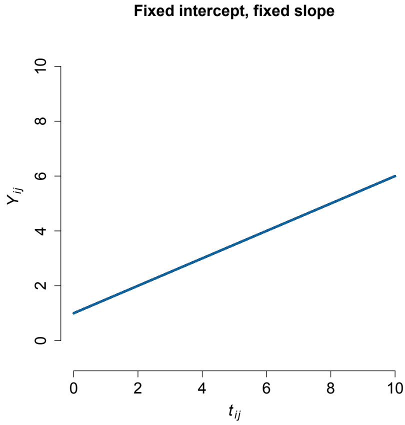
- **Random intercepts**

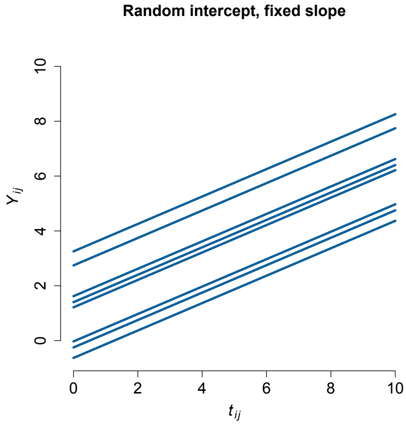
$$\begin{aligned} y_{ij} &= \beta_0 + \beta_1 t_{ij} + b_{i0} + \epsilon_{ij} \\ &= (\beta_0 + b_{i0}) + \beta_1 t_{ij} + \epsilon_{ij} \end{aligned}$$

- **Random intercepts and slopes**

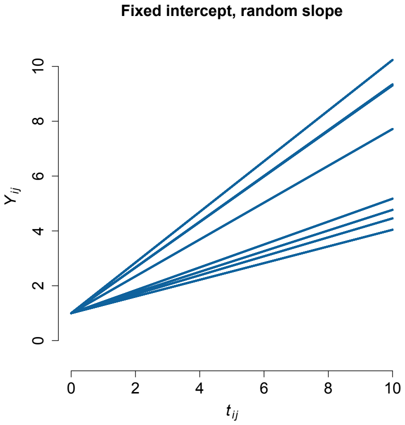
$$\begin{aligned} y_{ij} &= \beta_0 + \beta_1 t_{ij} + b_{i0} + b_{i1} t_{ij} + \epsilon_{ij} \\ &= (\beta_0 + b_{i0}) + (\beta_1 + b_{i1}) t_{ij} + \epsilon_{ij} \end{aligned}$$

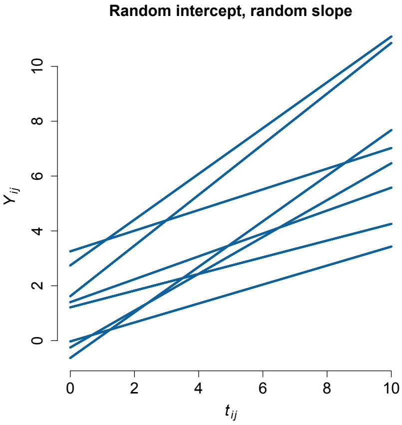
Choices for random effects





Choices for random effects





Choices for random effects: D

D quantifies random variation in trajectories across subjects

$$D = \begin{bmatrix} D_{11} & D_{12} \\ D_{21} & D_{22} \end{bmatrix}$$

- $\sqrt{D_{11}}$ is the typical deviation in the level of the response
- $\sqrt{D_{22}}$ is the typical deviation in the change in the response
- D_{12} is the covariance between subject-specific intercepts and slopes
  - $D_{12} = 0$ indicates subject-specific intercepts and slopes are uncorrelated
  - $D_{12} > 0$ indicates subjects with high level have high rate of change
  - $D_{12} < 0$ indicates subjects with high level have low rate of change

$(D_{12} = D_{21})$

Induced correlation structure

What is the correlation between measurements on the same subject?

- **Random intercepts model**
  - Assuming $\text{Var}[\epsilon_{ij}] = \sigma^2$ and $\text{Cov}[\epsilon_{ij}, \epsilon_{ij'}] = 0$

$$
\begin{aligned}
y_{ij} &= \beta_0 + \beta_1 t_{ij} + b_{i0} + \epsilon_{ij} \\
y_{ij'} &= \beta_0 + \beta_1 t_{ij'} + b_{i0} + \epsilon_{ij'}
\end{aligned}
$$

$$
\begin{aligned}
\text{Var}[Y_{ij}] &= \text{Var}_b[\text{E}_Y(Y_{ij} \mid b_{i0})] + \text{E}_b[\text{Var}_Y(Y_{ij} \mid b_{i0})] \\
&= D_{11} + \sigma^2
\end{aligned}
$$

$$
\begin{aligned}
\text{Cov}[Y_{ij},\ Y_{ij'}] &= \text{Cov}_b[\text{E}_Y(Y_{ij} \mid b_{i0}), \text{E}_Y(Y_{ij'} \mid b_{i0})] \\
&\quad + \text{E}_b[\text{Cov}_Y(Y_{ij},\ Y_{ij'} \mid b_{i0})] \\
&= D_{11}
\end{aligned}
$$

Induced correlation structure

- **Random intercepts model** (continued)

$$\begin{aligned}
\text{Corr}[Y_{ij}, Y_{ij'}] &= \frac{D_{11}}{\sqrt{D_{11}+\sigma^2}\sqrt{D_{11}+\sigma^2}} \\
&= \frac{D_{11}}{D_{11}+\sigma^2} \\
&= \frac{\text{`Between'}}{\text{`Between'} + \text{`Within'}} \\
&\geq 0 \text{ (and } \leq 1)
\end{aligned}$$

  - Any two measurements on the same subject have the same correlation; does not depend on time nor the distance between measurements
  - Longitudinal correlation is constrained to be positive ($D_{11} \geq 0$, $\sigma^2 \geq 0$)

Induced correlation structure

- **Random intercepts and slopes model**
  - Assuming $\text{Var}[\epsilon_{ij}] = \sigma^2$ and $\text{Cov}[\epsilon_{ij}, \epsilon_{ij'}] = 0$

$$
\begin{aligned}
y_{ij} &= (\beta_0 + \beta_1 t_{ij}) + (b_{i0} + b_{i1} t_{ij}) + \epsilon_{ij} \\
y_{ij'} &= (\beta_0 + \beta_1 t_{ij'}) + (b_{i0} + b_{i1} t_{ij'}) + \epsilon_{ij'}
\end{aligned}
$$

$$
\begin{aligned}
\text{Var}[Y_{ij}] &= \text{Var}_b[\text{E}_Y(Y_{ij} \mid b_i)] + \text{E}_b[\text{Var}_Y(Y_{ij} \mid b_i)] \\
&= D_{11} + 2D_{12} t_{ij} + D_{22} t_{ij}^2 + \sigma^2
\end{aligned}
$$

$$
\begin{aligned}
\text{Cov}[Y_{ij}, Y_{ij'}] &= \text{Cov}_b[\text{E}_Y(Y_{ij} \mid b_i), \text{E}_Y(Y_{ij'} \mid b_i)] \\
&\quad + \text{E}_b[\text{Cov}_Y(Y_{ij}, \, Y_{ij'} \mid b_i)] \\
&= D_{11} + D_{12}(t_{ij} + t_{ij'}) + D_{22} t_{ij} t_{ij'}
\end{aligned}
$$

Induced correlation structure

- **Random intercepts and slopes model** (continued)

$$
\begin{aligned}
&\text{Corr}[Y_{ij}, Y_{ij'}] \\
&= \frac{D_{11} + D_{12}(t_{ij} + t_{ij'}) + D_{22} t_{ij} t_{ij'}}{\sqrt{D_{11} + 2D_{12} t_{ij} + D_{22} t_{ij}^2 + \sigma^2}\sqrt{D_{11} + 2D_{12} t_{ij'} + D_{22} t_{ij'}^2 + \sigma^2}}
\end{aligned}
$$

  - Any two measurements on the same subject may not have the same correlation; depends on the specific observation times

Likelihood-based estimation of β

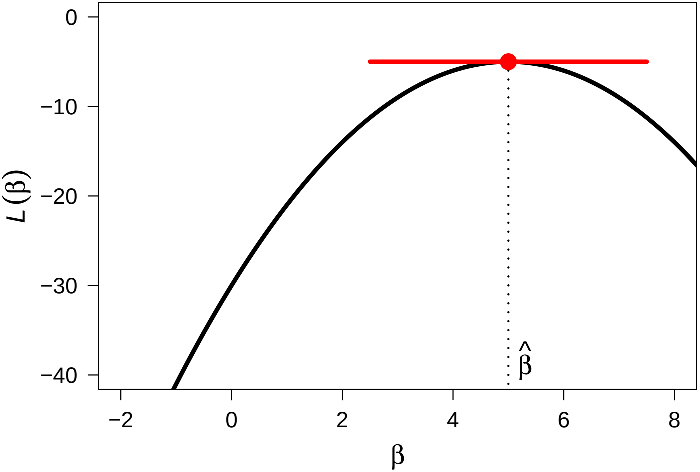
$\boxed{\text{Requires specification of a complete probability distribution for the data}}$

- Likelihood-based methods are designed for fixed effects, so integrate over the assumed distribution for the random effects

$$\mathcal{L}(\beta, \sigma, D) = \prod_{i=1}^{n} \int f_Y(y_i \mid b_i, \beta, \sigma) \times f_b(b_i \mid D) db_i$$

  where f_b is typically the density function of a Normal random variable
    - For linear models the required integration is straightforward because y_i and b_i are both normally distributed (easy to program)
    - For non-linear models the integration is difficult and requires either approximation or numerical techniques (hard to program)
    - Conditional likelihood methods treat the random effects as fixed and condition on statistics for them

Likelihood-based estimation of β



Likelihood-based inference for β

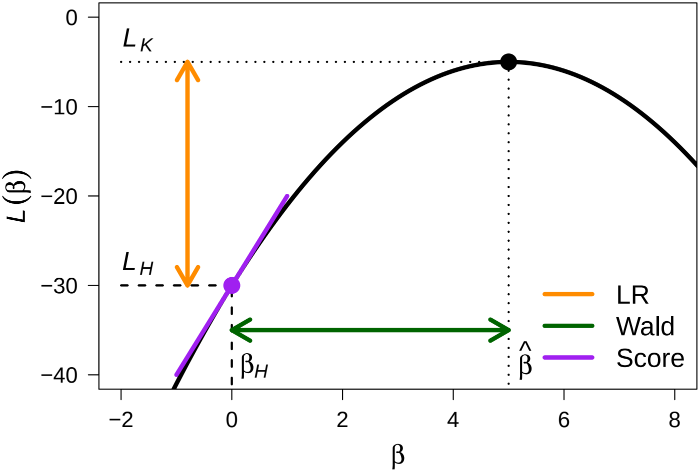
Consider testing fixed effects in nested linear mixed-effects models

$$H\colon\ \beta = \begin{bmatrix} \beta_1 \\ 0 \end{bmatrix} \quad \text{versus} \quad K\colon\ \beta = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix},$$

i.e., $H\colon\ \beta_2 = 0$

- Likelihood ratio test is valid with maximum likelihood estimation
  - Requires computation under the null and alternative hypotheses
- Likelihood ratio test may not be valid with other estimation methods
- Wald test (based on coefficient and standard error) is generally valid

Likelihood-based inference for β



Likelihood-based inference for D

Consider testing whether a random intercept model is adequate

$$H\colon\ D = \begin{bmatrix} D_{11} & 0 \\ 0 & 0 \end{bmatrix} \quad \text{versus} \quad K\colon\ D = \begin{bmatrix} D_{11} & \\ D_{12} & D_{22} \end{bmatrix},$$

i.e., $H\colon\ D_{12} = D_{22} = 0$

- Adequate covariance modeling is useful for the interpretation of the random variation in the data
- Over-parameterization of the covariance structure leads to inefficient estimation of fixed-effects parameters β
- Covariance model choice determines the standard error estimates for $\hat{\beta}$; correct model is required for correct standard error estimates
- Generally recommend against this inferential procedure
    - Specification for the covariance structure should be guided by *a priori* scientific knowledge and exploratory data analysis

Assumptions

Valid inference from a linear mixed-effects model relies on

- **Mean model**: As with any regression model for an average outcome, need to correctly specify the functional form of $x_{ij}\beta$ (here also $z_{ij}b_i$)
  - Included important covariates in the model
  - Correctly specified any transformations or interactions
- **Covariance model**: Correct covariance model (random-effects specification) is required for correct standard error estimates for $\hat{\beta}$
- **Normality**: Normality of ϵ_{ij} and b_i is required for normal likelihood function to be the correct likelihood function for y_{ij}
- n sufficiently large for **asymptotic inference** to be valid

⋆ These assumptions must be verified to evaluate any fitted model

Summary

- Mixed-effects models assume that each subject has a regression model characterized by subject-specific parameters; a combination of
  - Fixed-effects parameters common to all individuals in the population
  - Random-effects parameters unique to each individual subject
- Estimation and inference can focus both on average outcome levels and trends, and on heterogeneity across subjects in levels and trends
- Subject-specific random effects induce a correlation structure
- Parametric likelihood approach permits use of likelihood ratio test, but requires several assumptions that must be verified in practice

Issues

- Interpretation depends on outcomes and random-effects specification
- GLMM requires that any missing data are missing at random
- Issues arise with time-dependent exposures and covariance weighting

Key points

- Conditional mean regression model
- Model for population heterogeneity
- Subject-specific random effects induce a correlation structure
- Multiple sources of positive correlation
- Fully parametric model based on exponential family density
- Estimates obtained from likelihood function
- Conditional (fixed effects) and maximum (random effects) likelihood
- Approximation or numerical integration to integrate out b_i
- Requires correct parametric model specification
- Testing with likelihood ratio and Wald tests
- Conditional or subject-specific inference
- Induced marginal mean structure and 'attenuation'
- Missing at random (MAR)
- Time-dependent covariates and endogeneity
- R package `lme4`

Big picture

- Provide valid estimates and standard errors for regression parameters only under stringent model assumptions that must be verified $(-)$
- Provide population-averaged or subject-specific inference depending on the outcome distribution and specified random effects $(+/-)$
- Accommodate multiple sources of correlation $(+/-)$
- Require that any missing data are missing at random $(-/+)$